

Open Source hardware – A quick definition

Just like its software counterpart, an Open Source hardware device is available for anyone to copy, modify, manufacture and sell. There are no royalties or payments associated with it. The design's origins must be acknowledged by anyone using it and should allow anyone else using it the same rights.

Let's now look at the various 3D printers in the open source category.

1. RepRap 3D printer

The 'Replicating Rapid Prototyper' or 'RepRap' is one of the most widely used desktop 3D printers that prints plastic objects. Many of the parts of the RepRap printer are made of plastic, and it can print a duplicate version of itself (the plastic components). The version 'Mendel' is named after Gregor Mendel, the father of modern genetics. It uses the Fused Filament Fabrication (FFF) technique (see Chapter 2) to print 3D components. The total cost of making the Mendel, including the electronics and motor, is around \$520. It weighs around 7 kgs and can print at the speed of 15.0 cm³ per hour. The printing material used by the Mendel is PLA (a biodegradable polymer), ABS (a commonly used thermoplastic) and HDPE (a commonly used plastic).

The RepRap 3D printer was conceived as a complete printing solution rather than just a piece of hardware and so, to this end it comes bundled with the open source RepRap software with which you can design parts and print them using the printer. In fact, The RepRap printer uses the

